

Supporting Information — Robust Sameness Scoring Using a Shift-Tolerant Correlation Coefficient

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Software (STCC, experimental Local Contrast and the three reference methods), all benchmark harnesses, tabulated scores and the test suite are freely available under the MIT license at https://github.com/deepnmr/spectra_similarity_tool. Both STCC variants are implemented in `hsqc_lcc.py`.

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S1 Setup and preprocessing

Two spectra $a \in \{x, y\}$ are compared inside a common window $\Omega = [\delta_{\text{lo}}^{F2}, \delta_{\text{hi}}^{F2}] \times [\delta_{\text{lo}}^{F1}, \delta_{\text{hi}}^{F1}]$ (direct dimension $F2$, e.g. ^1H ; indirect dimension $F1$, e.g. ^{15}N or ^{13}C). Each processed point p has shifts $\delta_{F2}(p), \delta_{F1}(p)$ and intensity $I_a(p)$. With the default `clip` baseline, negative intensities are zeroed and every point is weighted by its local integration area so that spectra of different digital resolution integrate consistently:

$$w_a(p) = \max(I_a(p), 0) \cdot |\Delta\delta_{F1}(p)| \cdot |\Delta\delta_{F2}(p)|. \quad (\text{S1})$$

Negative peaks (edited HSQC). `clip` is correct for a non-edited spectrum, where negatives are baseline noise. In a **multiplicity-edited** HSQC, however, the sign encodes multiplicity (CH/CH_3

positive, CH₂ negative), not absence [1] — `clip` would silently delete every CH₂ crosspeak. For such data use `baseline="abs"`, which weights by $|I_a(p)|$ (replacing $\max(I_a, 0)$ in Eq. S1) and preserves the CH₂ peaks; all methods and the CLI expose this option. The sparse ¹H-¹³C peak-list benchmark already renders with $|intensity|$ for exactly this reason (§S9).

This is exactly the area weighting of the bin method [2]. The mass is normalized to $\sum_p w_a(p) = 1$; the final score is invariant to this scale, so normalization only guards against numerical overflow.

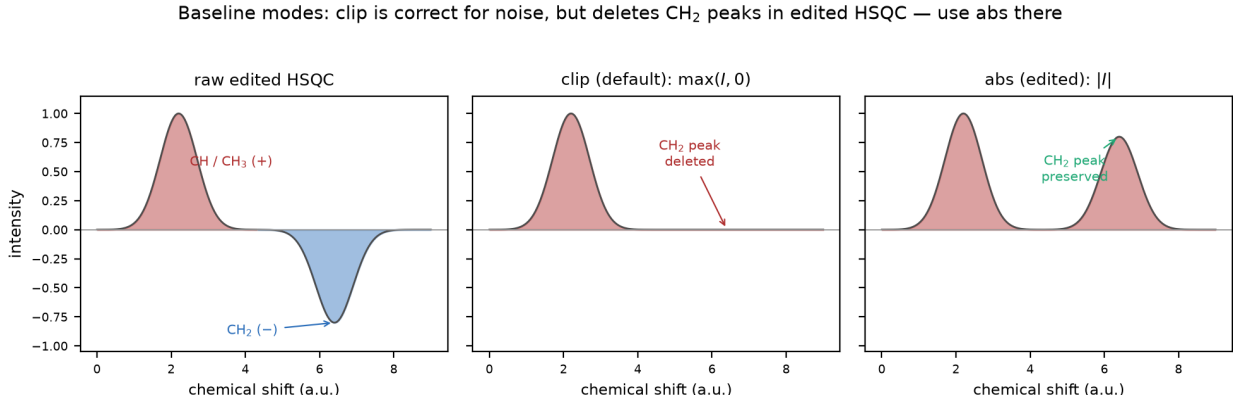


Figure S1. Baseline modes on a multiplicity-edited HSQC cross-section. **Left:** the raw signal, with a positive CH/CH₃ peak and a **negative** CH₂ peak (the sign encodes multiplicity, not absence). **Middle (clip, default):** $\max(I, 0)$ zeroes negatives — correct when negatives are baseline noise, but it silently **deletes the CH₂ crosspeak**. **Right (abs):** $|I|$ preserves the CH₂ peak. Use `clip` for a non-edited spectrum and `abs` for edited data; the sparse ¹H-¹³C benchmark renders with $|intensity|$ for exactly this reason (§S9).

S2 Rendering to a shared grid

The window is discretized once into a fixed grid shared by **both** spectra: K bins along $F1$ of width Δ_1 (default `step_f1` = 0.10 ppm) and L bins along $F2$ of width Δ_2 (default `step_f2` = 0.01 ppm). Each spectrum is rendered by summing its weighted points into cells (k, l) :

$$R_a[k, l] = \sum_{p: \delta(p) \in \text{cell}(k, l)} w_a(p), \quad k = 1, \dots, K, \quad l = 1, \dots, L. \quad (\text{S2})$$

Rendering both spectra onto the *same* edges is what makes the two images directly comparable regardless of their native resolutions [3]. **Optional STCC intensity compression:** to stop a single dominant amide from dominating the correlation, the rendered intensities may be raised to a power ρ (`--p`, default $\rho = 1$, i.e. off): $R_a[k, l] \leftarrow R_a[k, l]^\rho$. Both the unit-mass normalization and this optional compression are the standard scaling/normalization operations of quantitative NMR profiling [4,5]. The experimental Local-Contrast method does not use `--p`; its fixed square-root transform is applied after the lineshape blur (§S4).

S3 Lineshape blur

The discrete image is convolved with a separable Gaussian whose width is the physical NMR linewidth *plus* the expected chemical-shift drift, one σ per axis:

Rendering onto shared edges makes two spectra comparable regardless of native resolution

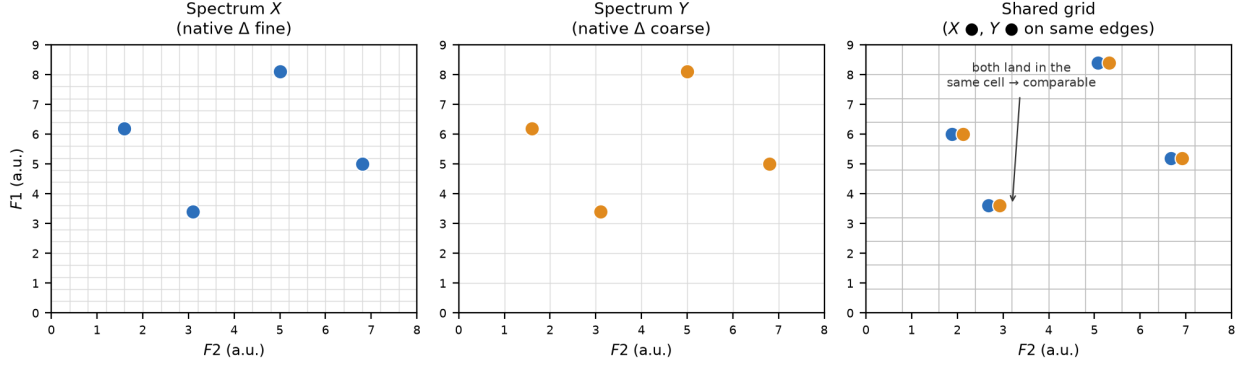


Figure S2. Rendering onto a shared grid. The **same** four crosspeaks are recorded by two instruments at different native digital resolutions (**left**, fine Δ ; **middle**, coarse Δ). Histogramming both onto one **shared** grid with identical cell edges (**right**) places corresponding peaks in the **same cell**, so the two images become directly comparable regardless of their native resolutions (Eq. S2).

$$\sigma_{F2} = \sqrt{\ell_{F2}^2 + d_{F2}^2}, \quad \sigma_{F1} = \sqrt{\ell_{F1}^2 + d_{F1}^2}, \quad (\text{S3})$$

with linewidth ℓ and expected drift d (defaults $\sigma_{F2} = 0.03$, $\sigma_{F1} = 0.30$ ppm for $^1\text{H}-^{15}\text{N}$). The 1-D kernel, with $s = \sigma/\Delta$ the width in pixels and radius $r = \lceil 3s \rceil$, is

$$g_{\sigma}(m) = \frac{\exp(-m^2/2s^2)}{\sum_{m'=-r}^r \exp(-m'^2/2s^2)}, \quad m = -r, \dots, r, \quad (\text{S4})$$

and the blurred (“lineshape-rendered”) image is the separable convolution

$$G_a = g_{\sigma_{F1}} *_{F1} (g_{\sigma_{F2}} *_{F2} R_a). \quad (\text{S5})$$

The continuous Gaussian is precisely what replaces the bin method’s *hard* bin edges with a *soft* response (see §S6).

S4 Mean-centring, the STCC score and experimental Local Contrast

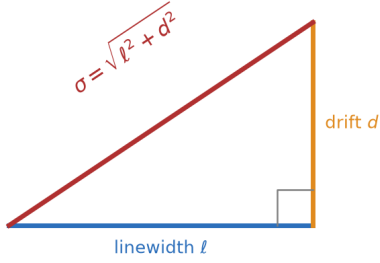
Each image is centred by subtracting its global mean

$$\bar{G}_a = \frac{1}{KL} \sum_{k,l} G_a[k,l], \quad \widehat{G}_a[k,l] = G_a[k,l] - \bar{G}_a, \quad (\text{S6})$$

and the similarity is the mean-centred normalized cross-correlation at **zero lag** (the Pearson correlation coefficient of the two flattened images), clamped to $[0, 1]$:

Lineshape blur σ = physical linewidth and expected drift added in quadrature

(a) Blur width combines two physical widths



(b) The blur sets the shift-tolerance scale

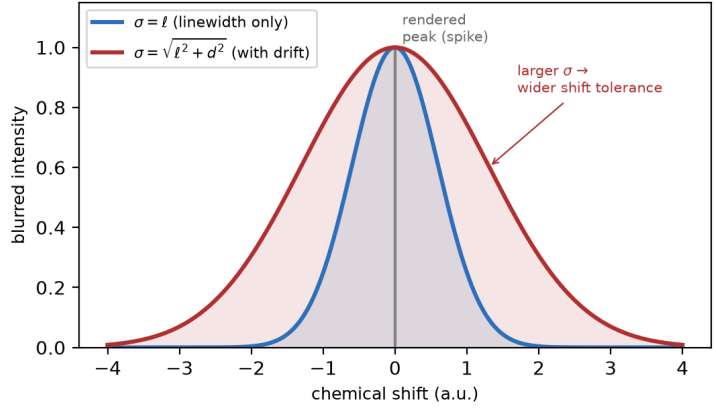


Figure S3. The lineshape blur width. **(a)** Each axis blur σ combines the physical NMR linewidth ℓ and the expected chemical-shift drift d in **quadrature**, $\sigma = \sqrt{\ell^2 + d^2}$ (Eq. S3). **(b)** Convolving a rendered peak (a spike) with this Gaussian sets the shift-tolerance scale: a larger σ (linewidth *plus* drift) tolerates a larger displacement than the linewidth alone would.

$$S(x, y) = \max \left(0, \frac{\sum_{k,l} \widehat{G}_x[k, l] \widehat{G}_y[k, l]}{\sqrt{\sum_{k,l} \widehat{G}_x[k, l]^2} \sqrt{\sum_{k,l} \widehat{G}_y[k, l]^2}} \right) \quad (\text{S7})$$

No shift search / alignment is performed — this is deliberate (§S6c).

Experimental Local-Contrast feature

The experimental candidate reuses the completed STCC render G_a from Eq. S5. It compresses the dynamic range, estimates a broad local background with a Gaussian three times wider than the STCC lineshape blur on each axis, and subtracts that background:

$$H_a = \sqrt{\max(G_a, 0)}, \quad B_a = g_{3\sigma_{F1}} *_{F1} (g_{3\sigma_{F2}} *_{F2} H_a), \quad F_a = H_a - B_a. \quad (\text{SLC1})$$

The Local-Contrast similarity is the un-centred cosine of the two signed feature images at zero lag, clipped to $[0, 1]$:

$$S_{\text{LC}}(x, y) = \text{clip} \left(\frac{\langle F_x, F_y \rangle}{\|F_x\| \|F_y\|}, 0, 1 \right). \quad (\text{SLC2})$$

If either feature norm is zero, the score is 0. The background factor 3 and square-root transform are fixed by design and are not new tuning options. The candidate preserves the same fixed ppm window, physical STCC blur, grid and zero-lag comparison; it adds no peak picking, alignment, dependency or fitted model. Square-root intensity weighting has precedent in spectral-library search

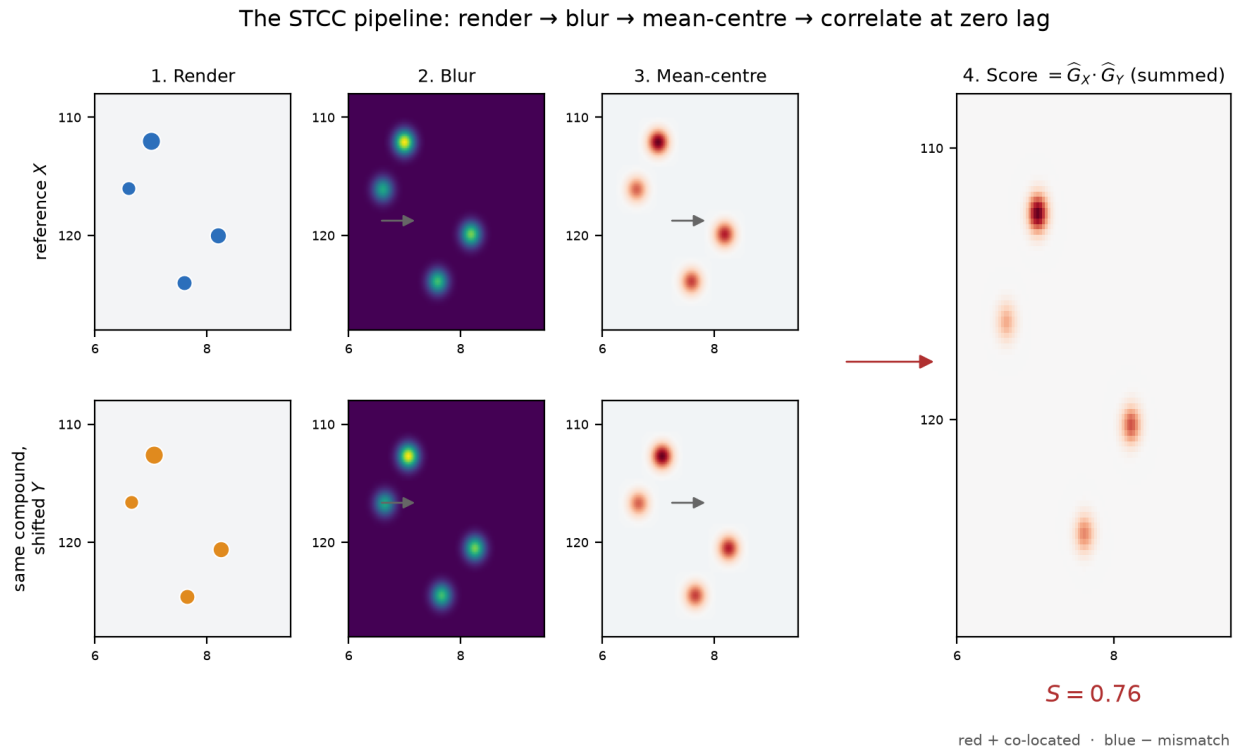


Figure S4. The four STCC steps end to end, on a toy pair. A reference X and a same-compound copy Y with a small titration drift are each (1) rendered as area-weighted points on a single **shared grid**, (2) blurred by the physical linewidth (a separable Gaussian), and (3) mean-centred (red = above the image mean, blue = below). (4) The similarity is the sum of the elementwise product $\widehat{G}_X \cdot \widehat{G}_Y$ (normalized): co-located intensity gives a **positive** (red) product and drives the score up, so this same-compound pair scores $S = 0.76$. No shift search is performed — the images are correlated at zero lag.

[38], but that precedent is a design rationale rather than direct evidence of NMR performance; the evidence here is restricted to the two repository benchmarks.

S5 Properties

Self-similarity is exactly 1. For $y = x$ the numerator and each factor of the denominator are the same sum $\|\widehat{G}_x\|^2$, so

$$S(x, x) = \frac{\|\widehat{G}_x\|^2}{\sqrt{\|\widehat{G}_x\|^2} \sqrt{\|\widehat{G}_x\|^2}} = 1, \quad (\text{S8})$$

and the $\min(1, \cdot)$ clamp in the code absorbs the last-ULP floating-point wobble so an identical spectrum returns 1.0 bit-exactly.

Symmetry. $S(x, y) = S(y, x)$ exactly, since the score is a symmetric inner product of the two centred images.

Range. As a Pearson coefficient $S \in [-1, 1]$ before clamping; the clamp maps anticorrelation to 0, giving $S \in [0, 1]$.

Monotonicity in drift. For an isolated peak on a grid large enough that mean-centring is negligible, S is a strictly decreasing function of the displacement magnitude (Eq. S9); this is checked numerically in the test suite.

Degenerate guard. A flat image ($\widehat{G}_a \equiv 0$) has zero denominator and returns 0; the windowing step already excludes empty windows upstream.

Local-Contrast contract. For non-degenerate features, Eq. SLC2 gives exact self-similarity, symmetry and a score in $[0, 1]$. Its normalization makes the score invariant to a global intensity multiplier. The tests additionally verify monotone decline under increasing chemical-shift drift, related-above-unrelated ordering after broad baseline and noise are added, a score change below 10^{-3} when each render step is halved, and explicit errors for non-finite data or non-positive finite blur/step values. A zero feature norm returns 0 as stated above.

S6 Why STCC beats the existing methods

The three source methods fail in *opposite* ways: the bin method [2] is shift-**brittle** (a peak straddling a bin edge splits its mass, so a small titration drift drops the score in a discontinuous jump), while the tree [6,7] and nearest-neighbour (NN) [8] methods are shift-**blind** (in a dense fingerprint every peak has a near neighbour, so both saturate near 1). STCC sits between them via three mechanisms.

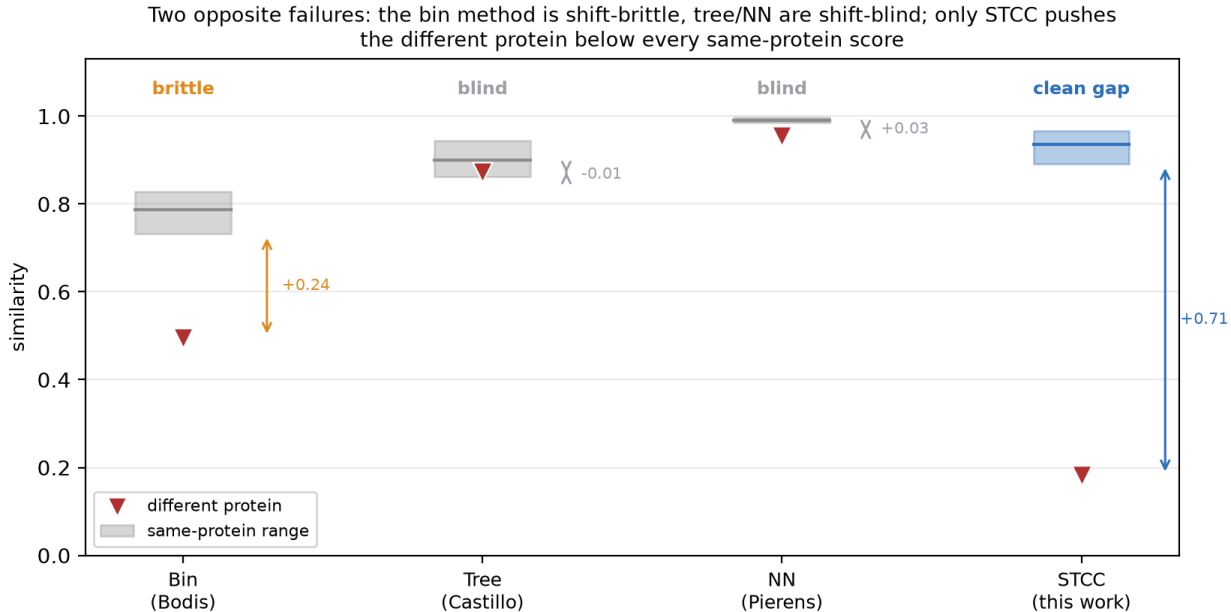


Figure S5. The two opposite failure modes, on the dense ^1H – ^{15}N benchmark (Table S2). For each method the shaded **band** spans the same-protein similarities (the titration series) and the red marker is the **different protein**. The bin method is shift-**brittle** — a physical titration drift erodes its same-protein scores down to 0.73–0.83, the lowest band shown. The tree and NN methods are shift-**blind** — the different protein sits *inside* (tree) or right at (NN) the same-protein band, so it is not separated. Only STCC pushes the different protein (0.18) below *every* same-protein score, opening a wide clean gap.

(a) Graded shift tolerance. Model a peak as a point mass; after the blur it is a Gaussian of width σ . Two copies of one peak displaced by $(\Delta_{F2}, \Delta_{F1})$ contribute to the centred, normalized correlation a factor

$$\exp\left(-\frac{\Delta_{F2}^2}{4\sigma_{F2}^2} - \frac{\Delta_{F1}^2}{4\sigma_{F1}^2}\right), \quad (\text{S9})$$

which is **smooth and monotone** in the drift — no bin-edge discontinuity. A physical titration drift ($\Delta_{1\text{H}} \sim 0.01\text{--}0.05$, $\Delta_{15\text{N}} \sim 0.2\text{--}0.8$ ppm) is a fraction of σ and costs little; a random relocation ($\Delta \gg \sigma$) drives the factor toward 0. The single knob σ sets the tolerance scale (main-text Figure 1b).

Graded shift tolerance: STCC decays smoothly where the bin method jumps

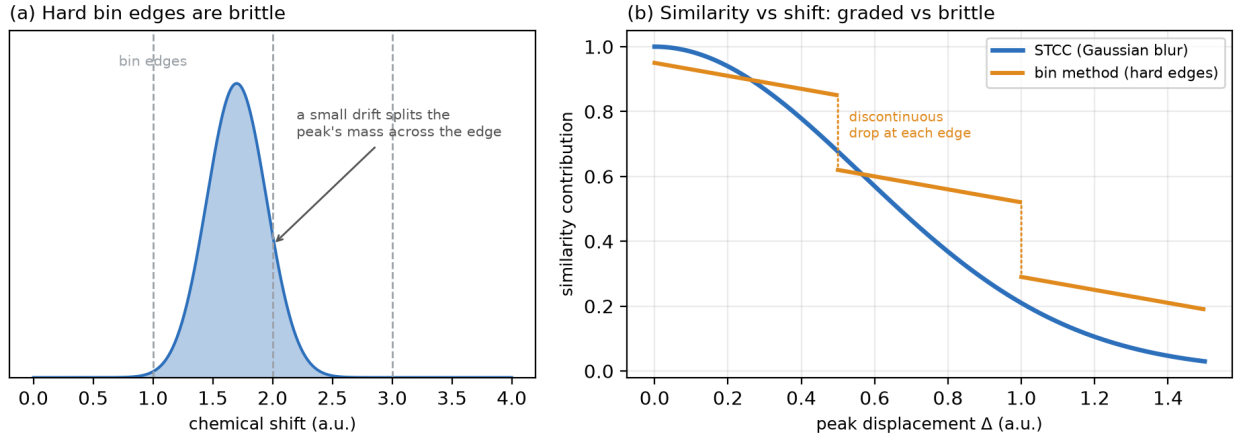


Figure S6. Graded vs brittle response to a shift. **(a)** The bin method’s edges are hard: as a peak drifts, a small displacement splits its mass across a bin boundary. **(b)** Similarity contribution vs peak displacement Δ . The Gaussian blur makes STCC decay **smoothly and monotonically** ($\exp(-\Delta^2/4\sigma^2)$, blue), so a physical drift costs little; the bin method (orange) instead **jumps discontinuously** each time the peak crosses an edge — a small drift can cause a sudden score drop. This is why the bin method is shift-brittle and STCC is not.

(b) Coverage penalty from mean-centring. Writing the score as a covariance,

$$S \propto \text{cov}(G_x, G_y) = \frac{1}{KL} \sum_{k,l} (G_x[k, l] - \bar{G}_x)(G_y[k, l] - \bar{G}_y), \quad (\text{S10})$$

a cell where one spectrum has a peak (value above its mean) and the other is empty (value below its mean) contributes a **negative** product and *lowers* S . Intensity that is not co-located is therefore actively penalized — a differently-scattered protein cannot hide behind coincidental proximity the way it does under NN’s one-way nearest-neighbour distance. This mean-centring step is worth $\approx +0.16$ of separation over an un-centred overlap on the protein benchmark (§S10).

(c) Zero lag on purpose. Adding a global shift search (FFT / block matching), as used in spectral-alignment tools [9], would let a different protein slide into registration and re-saturate —

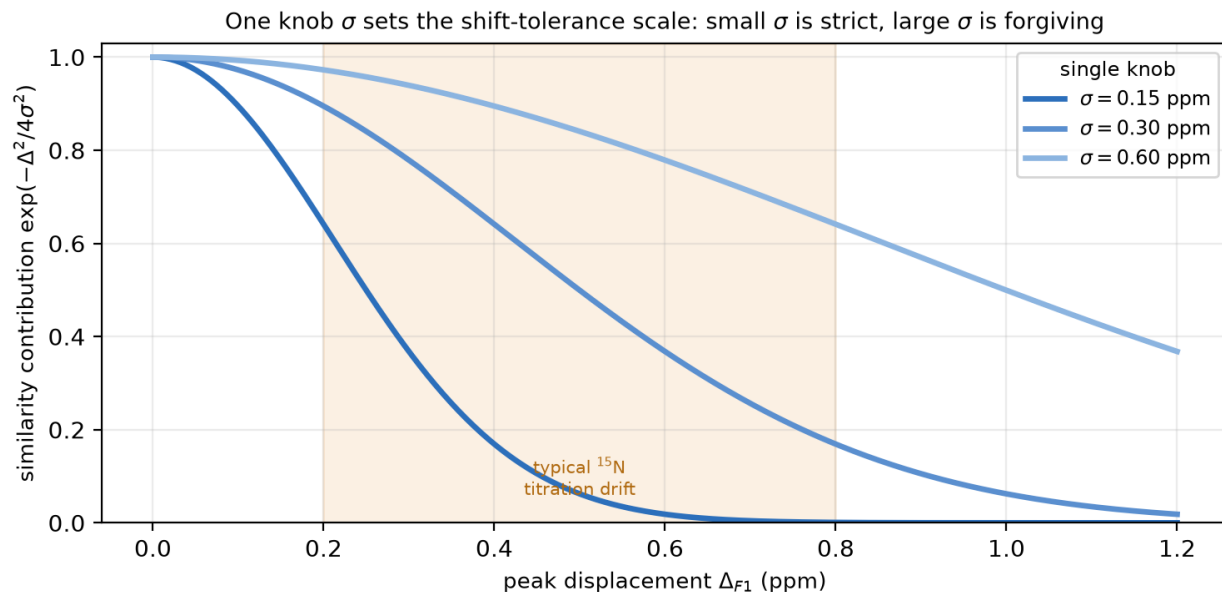


Figure S7. The single tuning knob. The same-peak contribution $\exp(-\Delta^2/4\sigma^2)$ (Eq. S9) versus peak displacement Δ , for three blur widths σ . One number sets the entire tolerance scale: a small σ is strict (a modest drift already costs similarity), a large σ is forgiving. The default is chosen so a typical ^{15}N titration drift (shaded) costs little while a random relocation drives the contribution toward zero.

the same failure as tree/NN. Scoring at zero lag keeps the position information that carries the discrimination.

S7 Parameters

Table S1. STCC parameters. The two blur widths are the only tuning levers, and both are set by spectroscopy: the linewidth sets the floor, the expected shift perturbation sets the tolerance.

symbol	flag	default (^1H - ^{15}N)	meaning
σ_{F2}	--sigma-f2	0.03 ppm	^1H linewidth + drift blur width (main $F2$ tolerance)
σ_{F1}	--sigma-f1	0.30 ppm	$^{15}\text{N}/^{13}\text{C}$ linewidth + drift (main tuning lever)
Δ_2	--step-f2	0.01 ppm	$F2$ render pixel width (≥ 3 px per σ ; score grid-invariant)
Δ_1	--step-f1	0.10 ppm	$F1$ render pixel width

symbol	flag	default (^1H - ^{15}N)	meaning
ρ	<code>--p</code>	1.0	intensity compression exponent (off)

For default STCC, centring is **on** and shift search is **off**; `center=False` gives the un-centred cosine ablation (§S10). Select the experimental candidate with `--method local-contrast`; its square root and background factor 3 are fixed, and `--p` does not apply. Negatives are handled by the baseline mode (`clip` default; use `abs` for multiplicity-edited HSQC — §S1). The CLI default remains `--method lcc`. For the ^1H - ^{13}C benchmark the blur widths are scaled to that regime ($\sigma_{^1\text{H}} = 0.05$, $\sigma_{^{13}\text{C}} = 0.5$ ppm).

S8 Benchmark 1 — dense protein ^1H - ^{15}N

Design. Reference spectrum: PRL3 experiment 2, a ^1H - ^{15}N HSQC. It is compared with the **same protein plus ligand** (experiments 4–14, a titration series that should score high — the chemical-shift-perturbation and fragment-based-screening setting [10–13]) and with a **different protein** (OAA experiment 103, which should score low). This same-protein-versus-different-protein discrimination — high for the same fold, low for a different one — is a component operation of higher-order-structure (HOS) comparability workflows, for which 2D-HSQC fingerprinting is an established atomic-resolution readout (§S13); it does not by itself exercise the subtle originator-versus-biosimilar comparison, the harder intermediate-score task the main text leaves to future work. Window $F2$ 6.5–10 $\times$ $F1$ 105–130 ppm. Bin widths for the bin method `min_bin_width_f2` = 0.1, `min_bin_width_f1` = 1.0; STCC blur $\sigma_{F2} = 0.03$, $\sigma_{F1} = 0.30$ ppm. Separation $\Sigma = \bar{S}_{\text{same}} - S_{\text{diff}}$; margin = $\min_e S_{\text{same}} - S_{\text{diff}}$.

Table S2. Per-comparison similarity, dense protein ^1H - ^{15}N benchmark. All methods self-score exactly 1.

comparison	Bin (Bodis 2009)	Bin + 45°	Tree (Castillo 2013)	NN (Pierens 2012)	STCC
2 vs 2 (self)	1.0000	1.0000	1.0000	1.0000	1.0000
2 vs 4 (+B12)	0.8106	0.8385	0.9401	0.9925	0.9662
2 vs 6 (+NB91)	0.7739	0.7990	0.8619	0.9981	0.9594
2 vs 8 (+benz)	0.8274	0.8570	0.9443	0.9877	0.9485
2 vs 10 (+GB001)	0.8164	0.8555	0.9179	0.9848	0.8972
2 vs 12 (+PF1)	0.7637	0.7912	0.8760	0.9978	0.9558
2 vs 14 (+PF2)	0.7320	0.7654	0.8624	0.9851	0.8911
2 vs OAA	0.4949	0.5681	0.8721	0.9550	0.1815

comparison	Bin (Bodis 2009)	Bin + 45°	Tree (Castillo 2013)	NN (Pierens 2012)	STCC
separation	0.29	0.25	0.03	0.04	0.75
margin	0.24	0.20	-0.01	0.03	0.71

STCC separates same-protein from different-protein spectra $\approx 2.6\times$ better in separation (0.75 vs 0.29) than the previous best, pushing the different protein (0.18) below *every* same-protein score. The old bin method left the different protein (0.49) close to the worst same-protein point (0.73); STCC opens a wide gap.

Expanded benchmark — a second decoy protein. The benchmark above scores against a single decoy (OAA). To give the negative side a distribution over distinct folds — and to make the dense test self-contained — we added an independent PRL3 apo $\rightarrow$ olsalazine titration (500 μ M PRL3(4–159), 23 points from exp 4 to exp 48) and a second decoy protein, the kinase domain **EphB3**, all read as processed Bruker spectra in the same ^1H 6.5–10 $\times$ ^{15}N 105–130 window (`bench_nhsqc.py`). As the ligand is titrated the chemical-shift perturbation grows and STCC decays **smoothly** from 0.98 to a plateau near 0.82 (Table S4), the graded shift tolerance of §S6a acting on real titration data; both decoy proteins fall far below the entire titration (OAA 0.19, EphB3 0.41), so STCC’s margin (0.41) is positive across two distinct folds and the negative side is a distribution, not one spectrum. Mean-centring still earns the gap (STCC separation 0.57 vs un-centred cosine 0.46), and the tree and NN measures again saturate (margins 0.12, 0.02). Local Contrast suppresses the decoys further (0.02 and 0.14), raising separation to 0.67 and margin to 0.52. STCC’s absolute margin is smaller than the six-point single-decoy benchmark (0.71) because this test is harder on both axes — a longer titration reaching larger shifts (min same 0.82 vs 0.89) and a tougher decoy (EphB3 0.41 vs OAA 0.19). Local Contrast is reported separately as a post hoc experimental candidate rather than used to redefine the primary comparison.

Table S3. Expanded dense ^1H – ^{15}N benchmark: PRL3 apo–olsalazine titration (23 same-protein points) vs two decoy proteins (OAA, EphB3). Separation = $\bar{S}_{\text{same}} - \bar{S}_{\text{decoy}}$; margin = $\min_e S_{\text{same}} - \max_d S_{\text{decoy}}$. Every method self-scores 1.

method	mean same	min same	OAA	EphB3
Local Contrast (experimental)	0.752	0.661	0.022	0.138
Bin (Bodis 2009)	0.817	0.753	0.461	0.500
Bin + 45°	0.865	0.810	0.521	0.563
Quad-tree (Castillo 2013)	0.946	0.912	0.793	0.749
Nearest neighbour (Pierens 2012)	0.986	0.983	0.959	0.962
Cosine, un-centred (STCC ablation)	0.897	0.853	0.360	0.507
STCC (default)	0.871	0.820	0.195	0.414

Table S3 (continued). Discrimination summary for the expanded dense benchmark.

method	separation	margin
Local Contrast (experimental)	0.672	0.523
Bin (Bodis 2009)	0.337	0.253

method	separation	margin
Bin + 45°	0.323	0.247
Quad-tree (Castillo 2013)	0.176	0.119
Nearest neighbour (Pierens 2012)	0.026	0.021
Cosine, un-centred (STCC ablation)	0.464	0.346
STCC (default)	0.567	0.405

Table S4. Selected points along the PRL3–olsalazine titration for STCC and experimental Local Contrast (reference = apo exp 2). Across all 23 titration points, every score stays above both decoys: STCC has a worst same-protein score of 0.82 versus worst decoy 0.41; Local Contrast has 0.66 versus 0.14.

exp	STCC	Local Contrast
4	0.98	0.95
8	0.95	0.89
12	0.93	0.85
16	0.90	0.80
20	0.87	0.76
24	0.85	0.73
28	0.85	0.71
32	0.84	0.69
36	0.83	0.68
40	0.83	0.67
44	0.83	0.67
48	0.82	0.66

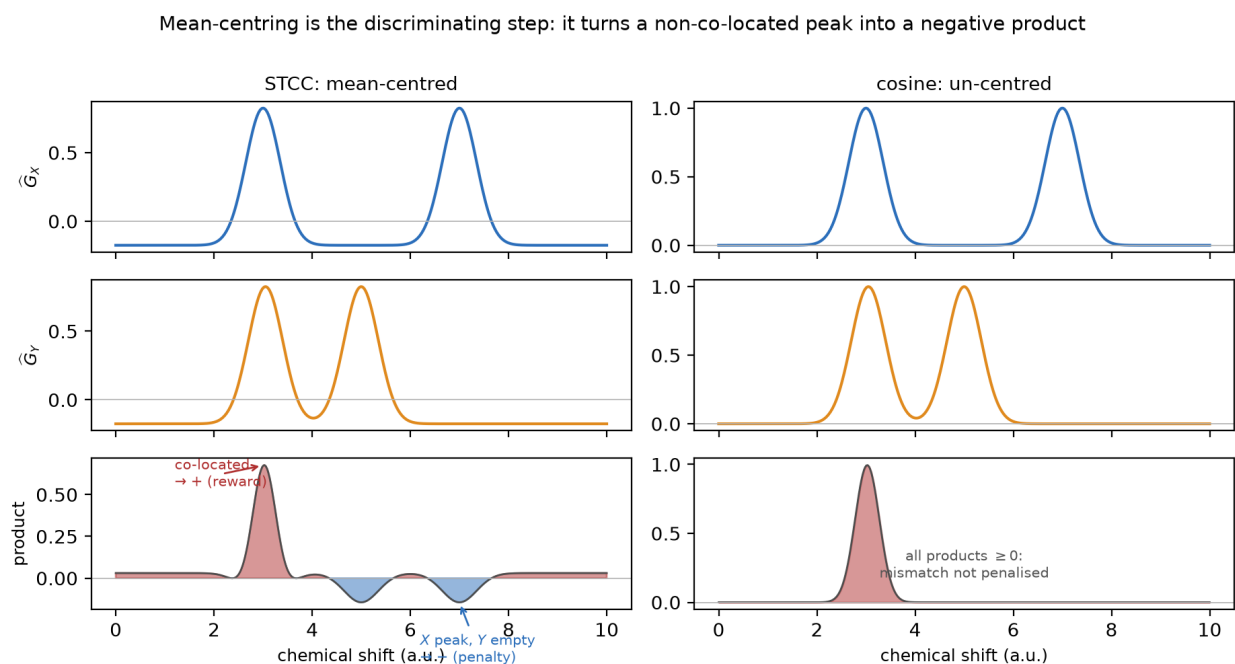


Figure S8. Why mean-centring is the discriminating step (1-D cross-section). Two images $\hat{G}_X$ (blue) and $\hat{G}_Y$ (orange) and their elementwise product (red = positive, blue = negative). **Left (STCC, mean-centred):** where both have a peak the product is **positive** (reward); where one has a peak and the other is empty, the empty image sits *below its mean* so the product is **negative** — non-co-located intensity is actively **penalized**. **Right (un-centred cosine):** every product is ≥ 0 , so a mismatched peak contributes nothing and cannot be penalized. Mean-centring is exactly what converts “overlap” into “discrimination”.

Scoring at zero lag keeps position information; a global shift search would slide different spectra into false registration

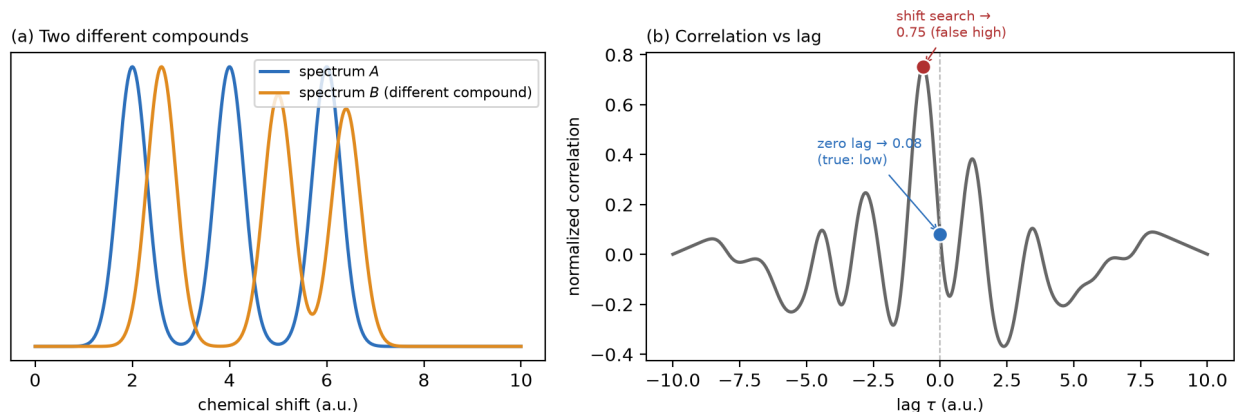


Figure S9. Why the score is read at **zero lag**. (a) Two genuinely different compounds (unequal peak spacing, so no rigid shift aligns them all). (b) Their normalized correlation as a function of a rigid lag τ . At zero lag the score is correctly **low** (0.08, blue); a global shift search would instead report the **maximum** over all lags (0.75, red), sliding the different spectra into a partial false registration — the same over-tolerant saturation as tree/NN. STCC scores only at $\tau = 0$, keeping the position information that carries the discrimination.

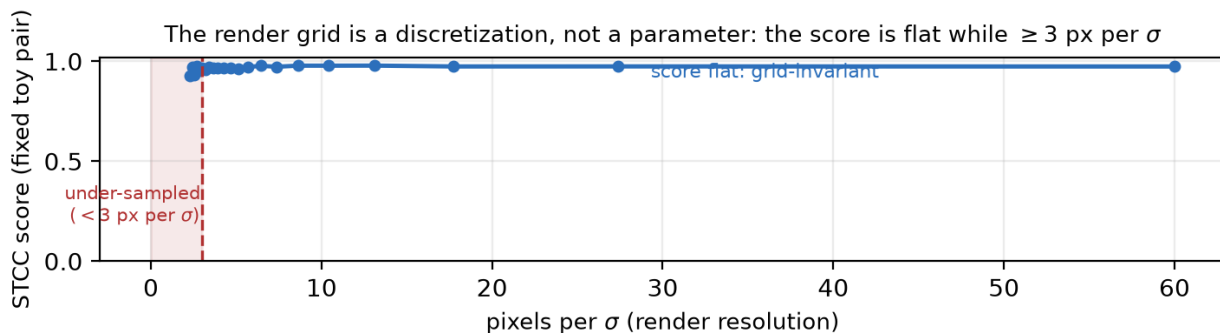


Figure S10. The render grid is a discretization, not a tuning parameter. A fixed same-compound toy pair is scored while the render pixel size is swept: the STCC score is **flat** while the Gaussian is sampled by ≥ 3 pixels per σ (red zone = under-sampled). The choice of grid does not affect the result.

S9 Benchmark 2 — sparse small-molecule ^1H - ^{13}C

Design. Public HSQC peak lists from the simpleNMR example set [14] (EricHughesABC/simpleNMR). Six nominal compounds, each recorded **twice** (a variant pair — e.g. menthol in two solvents). One nominal pair, **olivetol**, was excluded: its two source files (`Olivetol.json` and `Olivetol_A.json`) are **byte-identical** (same MD5, identical peak lists), so it was a self-comparison scoring 1.00 for every method — an information-free positive that inflated `mean_same`. The benchmark now uses the **5 genuinely distinct** same-compound pairs (10 spectra) and asks each measure to score them above the **40 different-compound pairs**. `bench_13c.py` raises at load time on any same-pair that renders to an identical image, so a duplicate cannot silently re-enter. Peak lists are rasterized to a synthetic 2D **stick** spectrum, using `|intensity|` so edited-HSQC CH_2 peaks are kept (§S1); each method then applies its own shift-tolerance processing. This avoids the former method-neutral pre-blur and the resulting double blur for STCC-family methods. Window ^1H 0–10 $\times$ ^{13}C 0–165 ppm; STCC blur $\sigma_{^1\text{H}} = 0.05$, $\sigma_{^{13}\text{C}} = 0.5$ ppm.

Table S5. Aggregate scores, sparse small-molecule ^1H - ^{13}C benchmark (5 same-compound pairs, 40 different). All methods self-score exactly 1. The un-centred cosine is STCC without mean-centring (§S6b), included here as the ablation baseline (bug-fix: it is now a reproducible benchmark row, not an asserted number).

method	mean same	mean diff	separation	margin
Local Contrast (experimental)	0.824	0.004	0.820	0.434
Bin (Bodis 2009)	0.748	0.073	0.675	0.212
Bin + 45°	0.768	0.096	0.672	0.194
Quad-tree (Castillo 2013)	0.914	0.502	0.413	0.069
Nearest neighbour (Pierens 2012)	0.997	0.898	0.099	0.041
Cosine, un-centred (STCC ablation)	0.747	0.002	0.744	0.373
STCC (default)	0.746	0.002	0.745	0.374

Table S6. Per-pair same-compound scores. Two pairs are genuinely shifted (menthol in two solvents; a rotenone re-measurement); on these the shift-tolerant tree/NN keep the same compound high while the position-sensitive bin/STCC penalize the shift. The catch is that tree/NN tolerate *everything*, which is why their different-compound scores (Table S5) are also high. The un-centred

cosine column is identical to STCC to two decimals — in this sparse regime mean-centring changes nothing (see the ablation, §S10).

pair	Bin	Tree	NN	Cosine	STCC	Local Contrast
menthol (two sol-vents)	0.734	0.967	0.988	0.421	0.421	0.467
rotenone (re-measured)	0.323	0.839	1.000	0.389	0.388	0.727
santonin	0.973	0.993	1.000	1.000	1.000	0.999
chartreusin	0.813	0.852	0.999	0.930	0.930	0.932
indanone	0.897	0.923	1.000	0.992	0.992	0.997

Over a large common window a different small molecule still supplies *some* peak near every peak and a similar mass-centre structure, so the NN distance and the tree overlap stay small — the same saturation mechanism as in the protein case.

S10 Robustness and ablation

Ablation — mean-centring is the discriminating step, in the dense regime. Replacing the mean-centred correlation (Eq. S7) with an un-centred cosine overlap on the *same* rendered/blurred images drops the dense-protein separation from **0.75 to 0.59**, confirming mean-centring (§S6b) as the operation that turns overlap into discrimination when the amide fingerprint is crowded. In the **sparse** ^1H - ^{13}C regime, by contrast, the un-centred cosine and centred STCC nearly coincide (separation **0.74445 vs 0.74466**; Table S5, Table S6): where crosspeaks rarely overlap there is almost no non-co-located intensity for mean-centring to penalize, so its benefit is specific to the dense fingerprint that is the hard case this measure targets. The cosine baseline is a real, reproducible benchmark row in both harnesses (`cosine_similarity` in `hsqc_lcc.py`), not an asserted number.

Robustness — the gain is not a tuned parameter. On the protein benchmark STCC beats the bin method across the whole physical blur range: separation is $\Sigma = 0.71$ – 0.77 at $\sigma = 0.02$ – 0.04 / 0.20 – 0.40 ppm, and still $\Sigma = 0.36$ (vs the bin method’s 0.29) when the grid is coarsened all the way to the bin method’s own resolution (0.10 / 1.0 ppm).

Experimental Local Contrast. On the expanded dense benchmark, the candidate increases separation/margin from STCC’s $0.567/0.405$ to $0.672/0.523$. On the stick-input sparse benchmark it increases $0.745/0.374$ to $0.820/0.434$, driven especially by the shifted rotenone pair ($0.388 \rightarrow 0.727$). These are composite results for square-root compression *together with* fixed-scale local background subtraction; the present experiment does not assign the gain to either component alone. Its representative runtime is $1.34\times$ STCC. The candidate remains experimental because the held-out operating metrics tie default STCC (§S13).

Table S7. Direct comparison of default STCC and its experimental Local-Contrast extension.

regime	method	mean same	mean different	separation	margin
Dense, 23 same + 2 decoys	STCC	0.871	0.305	0.567	0.405
	Local Contrast	0.752	0.080	0.672	0.523
Sparse, 5 same + 40 different	STCC	0.746	0.002	0.745	0.374
	Local Contrast	0.824	0.004	0.820	0.434

The former sparse blur-sweep table is omitted because it was produced from already Gaussian-blurred peak-list images. The current benchmark starts from sticks so that each method owns its smoothing; a future sweep must use that same preprocessing before it is compared with the current table.

Caveats. (i) In the sparse benchmark, spectra are **stick-rendered from peak lists**, not measured 2D matrices. Each method owns its smoothing, but exact values remain specific to these coordinates, intensities and grid; the *ranking* is the result, not a universal calibration. (ii) The tree and NN methods here are the repository’s representative reimplementations run inside one common wide window; the original papers tune per-spectrum windows and parameters, so this is not a verdict on those methods as their authors deployed them — only on how they behave as drop-in global similarity scores. (iii) The between-compound contrast is strong (40 pairs) but the same-compound side is only 5 pairs, of which two (menthol, rotenone) are the genuinely-shifted, discriminating tests; a byte-identical sixth pair (olivetol) was excluded (§S9).

S11 Relationship to the source methods

Table S8. STCC keeps the physically meaningful ingredient of each predecessor. Local Contrast is an experimental feature transform layered on the completed STCC render.

ingredient	taken from
window + negative handling (<code>clip/abs</code>) + area weighting; normalization to unit mass	Bodis et al. 2009 (bins)
physical blur width as shift tolerance ($\sigma = \sqrt{\ell^2 + d^2}$)	Castillo et al. 2013 (tree shift-insensitivity)
a peak / lineshape picture of the spectrum	Pierens et al. 2012 (NN peaks)
mean-centred zero-lag correlation as the discriminating step	this work (synthesis)
square root + fixed- 3σ local background subtraction	this work (experimental extension)

The mean-centred zero-lag correlation is itself the zero-mean normalized cross-correlation (ZNCC) standard in image registration and template matching; the contribution of this work is the *synthesis* — applying it to physically blurred 2D-HSQC images together with the ingredients above — not the correlation operation itself. The un-centred cosine of the same rendered/blurred images (`center=False`) is the natural ablation of the mean-centred-correlation row — it isolates what mean-centring buys (§S10) and is exactly the contrast-angle similarity of mass-spectral library search [15,16]. Local Contrast instead uses the signed feature in Eq. SLC1 with an un-centred cosine. The current benchmark tests that composite transform, not its two ingredients separately.

S12 Software, reproduction and test suite

Implementation. STCC and Local Contrast are pure NumPy [17] (`hsqc_lcc.py`); they read the processed Bruker data (the `2rr` matrix and its `procs` parameters) directly through the repository’s own reader — without a general-purpose package such as `nmrglue` [18] — need no peak picking and no external solver. On a representative dense pair, STCC runs in 12.8 ms and Local Contrast in 17.2 ms (1.34×; Apple Silicon, single core). The processed input can come from any standard NMR processing pipeline [19–22]. Figures use Matplotlib [23].

Local-Contrast JSON records the method, both blur widths, actual grid steps, grid shape, fixed ppm ranges, `background_factor=3` and `intensity_transform="sqrt"`, so the experimental calculation is self-describing. Non-finite spectra/ranges and non-positive or non-finite blur/step inputs raise explicit errors.

Reproduction. All benchmark harnesses are in the repository:

- `python3.11 bench.py` — dense protein ^1H - ^{15}N benchmark (Table S2), writing `results/method_comparison.{csv,json}` and `results/lcc_comparison.png`.
- `python3.11 bench_nhsqc.py` — expanded dense benchmark with two decoy proteins (Tables S3, S4; main-text Table 2), reading the `Nhsqc/` PRL3-olsalazine titration and OAA/EphB3 spectra and writing `results/nhsqc_dense.json`.
- `python3.11 bench_13c.py` — sparse small-molecule ^1H - ^{13}C benchmark (Tables S5, S6), downloading the simpleNMR peak lists on first run, rasterizing sticks and writing `results/comparison_13c.json`. It excludes the byte-identical olivetol pair and raises on any same-pair that renders identically.
- `python3.11 bench_retrieval.py` — top-1/MRR, AUROC/AUPRC, leave-one-compound-out threshold error, rejection FPR and compound-cluster bootstrap comparisons on the sparse regime (Table S9), writing `results/retrieval_13c.json`.
- `python3.11 results/make_plots.py` — regenerates Figure 2 (and `comparison_13c.png`) from the JSON; `python3.11 results/make_fig3.py` — regenerates Figure 3 (bootstrap CIs + ablation) from `retrieval_13c.json`; `python3.11 results/make_schematic.py` — regenerates Figure 1; `python3.11 results/make_si_figs.py` — regenerates the ten SI method figures (S1–S10).

Test suite. A 46-test suite (all passing) checks the properties proved in §S5 and the Local-Contrast contract: exact self-score, symmetry, $[0,1]$ range, monotonic drift response, global intensity-scale invariance, baseline/noise related-above-unrelated ordering, half-step grid invariance and explicit invalid-input errors. Existing regressions cover mean-centring, the `abs` baseline and the duplicate-pair guard. Run with `python3.11 -m pytest` (files under `tests/`: `test_hsqc_lcc.py`, `test_hsqc_methods.py`, `test_hsqc_similarity.py`, `test_bench_13c.py`, `test_bench_nhsqc.py`, `test_bench_retrieval.py`, `test_spectrum_similarity.py`).

S13 Statistical strength: bootstrap CIs and a retrieval test

These metrics run on the public sparse ^1H - ^{13}C data and are reproducible with `python3.11 bench_retrieval.py` (writes `results/retrieval_13c.json`).

Bootstrap confidence intervals and operating metrics. The sparse-regime separation is resampled 10^4 times at the level of the five **compounds** (whole compounds rather than the 45 pairs), so the dependence induced by each spectrum recurring in eight different-compound pairs is respected [24]. The cluster 95% intervals are wide: STCC 0.508–0.982 and Local Contrast 0.623–

0.985. A paired compound-cluster bootstrap, which evaluates both methods on the same resamples, gives Local Contrast – STCC $\Delta = 0.0757$ (95% CI 0.0014–0.2057, $P(\Delta \leq 0) = 0.0067$).

The promotion decision is based on held-out operation, not separation alone. For each held-out compound, a threshold minimizing balanced error is learned from the remaining compounds and then applied to every pair involving the held-out compound. Both STCC and Local Contrast give AUROC and AUPRC of 1.00, top-1 and MRR of 1.00, zero leave-one-compound-out error and zero rejection FPR and false-negative rate. The candidate therefore improves descriptive separation on these five compounds without improving the held-out decision boundary. It remains experimental.

Table S9. Current stick-input sparse ^1H – ^{13}C evaluation. The CI is the compound-cluster interval; top-1 and MRR [25] use the 10-spectrum retrieval library.

method	separation	cluster 95% CI	AUROC	AUPRC
STCC (default)	0.745	0.508–0.982	1.00	1.00
Local Contrast (experi- mental)	0.820	0.623–0.985	1.00	1.00

Table S9 (continued). Held-out and retrieval operating metrics.

method	LOO error	rejection FPR	top-1	MRR
STCC (default)	0.00	0.00	1.00	1.00
Local Contrast (experimental)	0.00	0.00	1.00	1.00

Retrieval (dereplication). Cast as a library search — for each of the 10 spectra, rank the other 9 by similarity and ask whether the true same-compound partner is retrieved first — **every** method achieves top-1 = 1.00 and MRR = 1.00. This is an honest null result: on a small, clean library the correct partner is always the single most-similar spectrum, even for the saturating tree/NN methods whose *different*-compound scores are also high. Retrieval accuracy therefore does **not** discriminate the methods at this scale; the discriminating quantity is the thresholded same/different decision (separation and margin), which is what would matter against a large decoy library or with a rejection threshold. The held-out threshold test above is also tied for STCC and Local Contrast; a larger, harder library is required to test whether the descriptive gap becomes an operating advantage. We report the null result rather than omit an experiment that did not favour the proposed method. A prospective, at-scale dereplication study [26,27] against a community library (BMRB [28], HMDB [29], NMRShiftDB [30]) is the natural way to turn this into a discriminating chemical application; STCC provides the scoring primitive, complementary to the multivariate workflows of NMR-based profiling [31]. A second, equally natural application is the **higher-order-structure (HOS) comparability** of protein biotherapeutics — biosimilar-versus- originator, or before/after a formulation or process change — where the fold is confirmed by matching 2D-HSQC fingerprints and where 2D-NMR is an established, high-precision readout [32–37]. The dense-protein benchmark (§S8) exercises the distinct-fold end of that task — cleanly separating fingerprints of different folds and tracking a titration smoothly — but not the subtle originator-versus-biosimilar discrimination, which falls in the intermediate-score regime and is left to future work (main text).

S14 Supporting references

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